

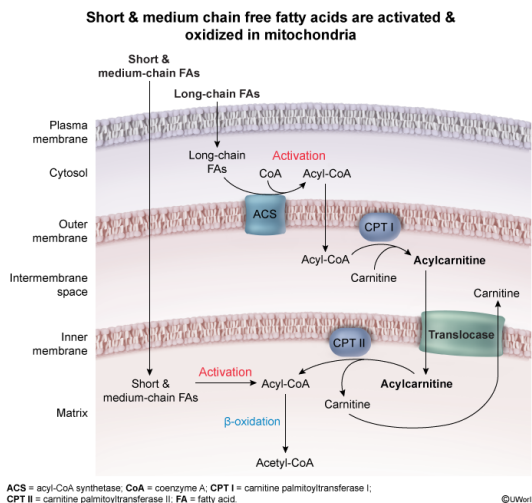
Where in the cell are short- and medium-chain fatty acids activated and oxidized?

- ☐ A. Cytosol [24%]
- ☐ B. Nucleus
- ☐ C. Smooth endoplasmic reticulum [14%]
- ✓ ☒ D. Mitochondria [60%]

Correct

61% answered correctly

Explanation:



Fatty acids (FAs) are hydrocarbon chains that contain a carboxylic acid group at one end. The hydrocarbon chains can consist entirely of single bonds, or they may contain one or more double bonds, giving rise to the classification of FAs as either **saturated or unsaturated**, respectively. FA chain length varies considerably and is also used to classify FAs: Short-chain FAs (SCFAs) have at most 5 carbons; medium-chain FAs (MCFAs) have 6-12 carbons; and long-chain FAs (LCFAs) have more than 12 carbons.

In a process collectively known as **FA oxidation** (or beta oxidation), FA chains are broken down into two-carbon **acetyl-coenzyme A (CoA) units** that can be used for ATP synthesis via the citric acid cycle. Before this can occur, the hydrocarbon chains of cytosolic FAs must enter the mitochondrial matrix. For SCFAs and MCFAs, this is accomplished via simple diffusion, but the hydrocarbon chain of LCFAs must be transiently linked to a molecule called carnitine to cross the inner mitochondrial membrane.

The question asks for the cellular location of SCFA and MCFA activation and oxidation. SCFAs and MCFAs are activated by acyl synthetase enzymes located in the **mitochondrial matrix**, thereby forming the "activated" acyl-CoA versions of the FAs. The original acyl-CoA versions of LCFAs are transiently converted to an acylcarnitine to cross the inner mitochondrial membrane before being converted back to an acyl-CoA in the matrix. **Within the matrix, fatty acyl-CoA molecules of all lengths undergo FA oxidation.**

(Choice A) FAs are **synthesized** in the cytosol. In addition, a long-chain acyl synthetase in the outer mitochondrial membrane converts cytosolic LCFAs to their original acyl-CoA versions (which are subsequently converted to long-chain acyl-CoAs).

(Choice B) Although FAs can enter nuclei and activate transcriptional responses, they are not oxidized there.

(Choice C) FAs are used in the synthesis of certain lipids by the smooth endoplasmic reticulum, but fat oxidation does not occur there.

Educational objective:

Fatty acid oxidation and activation to acyl-coenzyme A molecules are carried out by mitochondria.

Time spent:0 sec

Subject:
Biochemistry

QID:403237

System:
Metabolic Reactions